Program : **Diploma in Computer Application and Business Management**	
Course Code : **1257**	Course Title: **Introduction to Python**
Semester : **1**	Credits: **2**
Course Category: **Program Basic**	
Periods per week: **3 (L:1, T:0, P:2)**	Periods per semester: **45**

Course Objectives:

- To provide a hands-on experience in programming concepts using Python Programming Language. Students will be able to develop programs using arrays, functions, and files.
- To enhance the problem-solving skills of students.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Basic understanding of Computer terminologies		Introduction to Python	High School

Course Outcomes:

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Apply the fundamental Python syntax and semantics and be fluent in the use of Python control flow statements.	10	Applying
CO2	Develop proficiency in the handling of strings and functions.	12	Applying
CO3	Construct Python programs by utilizing the data structures like lists, dictionaries, tuples and sets.	10	Applying
CO4	Identify the commonly used operations involving file systems and exception handling	10	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Apply the fundamental Python syntax and semantics and be fluent in the use of Python control flow statements.**		
MO1.1	Develop simple Python programs to illustrate operator precedence	2	Applying
MO1.2	Develop simple Python programs using decision control structures	4	Applying
MO1.3	Implement Python Programs using loop control structures.	4	Applying
CO2	**Develop proficiency in the handling of strings and functions.**		
MO2.1	Develop Python program using functions (like swapping the value in two variables, to check whether a given number is odd or even, etc)	3	Applying
MO2.3	Python program to demonstrate default arguments	3	Applying
MO2.3	Implement Python programs using strings	6	Applying
	Lab Exam – I	1½	
CO3	**Construct Python programs by utilizing the data structures like lists, dictionaries, tuples and sets.**		
MO3.1	Implement Python programs using lists	3	Applying
MO3.2	Implement Python programs using dictionaries	3	Applying
MO3.3	Implement Python programs using lists	3	Applying
MO3.4	Implement Python programs using sets	1	Applying

CO4	**Identify the commonly used operations involving file systems and exception handling**		
MO4.1	Develop programs to illustrate exception handling	4	Applying
MO4.2	Implement programs to create, and to open and close a file	3	Applying
MO4.3	Develop programs to append a file and similar other functions.	3	Applying
	Lab Exam – II	1½	

Text / Reference:

T/R	Book Title/Author
R1	Gowrishankar S, Veena A, **Introduction to Python Programming**, 1st Edition, CRC Press/Taylor & Francis, 2018
R2	**Python3 for Absolute Beginners** by Tim Hall & J P Stacey
R3	Head First Python: A Brain-Friendly Guide by Paul Barry

Online Resources:

Sl.No	Website Link
1	https://www.w3schools.com/python/
2	https://www.tutorialspoint.com/python/
3	https://www.geeksforgeeks.org/python-programming-language/